

Interactive graph-based simulator for divisibility

Application example

THE NUMBER 895958 IS DIVISIBLE BY 7

This document presents, through a practical example, how to use the interactive graph-based simulator for the study of the divisibility of a number. The example used is the number 895958 and the modulus 7, with the aim of verifying, through a visual representation, that $895958 \pmod{7} = 0$ and, therefore, that 895958 is divisible by 7.

This resource complements the “Interactive graph-based activity for divisibility”, in which the user was required to carry out the procedure step by step and obtain the result. In this case, the simulator guides the user through the entire process, visually displaying the walk through the graph, the operations performed, and the corresponding calculations until the final solution is reached.

The purpose of this manual is to explain how the simulator works and to facilitate its use as a support resource for understanding, in a visual and interactive way, the procedure used to determine whether a number is divisible by another number.



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USER MANUAL

WHAT IS THIS RESOURCE?

This interactive resource allows users to determine whether a number is divisible by a given modulus (m) by means of a divisibility graph (or state graph). Each circle in the graph represents a possible remainder of a division: state 0 represents remainder 0; state 1 represents remainder 1; state 2 represents remainder 2; and so on up to ' $m-1$ '.

The resource processes the number digit by digit, using the following operation:

$$r_0 = 0; \quad r_i = (10 \cdot r_{i-1} + d) \bmod m \quad (i = 1, 2, \dots, m-1),$$

where ' r_{i-1} ' is the remainder obtained up to that point, ' r_i ' is the new remainder, and ' d ' is the digit currently being processed. At the end, if the final remainder is 0, the number is divisible by the modulus.

HOW THE RESOURCE WORKS

The resource mainly uses four controls:

1. Modulus: the user must enter a number between 2 and 20.
2. Number: the user must enter the number to be analyzed.
3. Speed: allows the speed of the automatic animation to be adjusted. A higher speed makes the steps play more quickly.
4. Manual slider: allows the process to be controlled step by step.

The main element on the screen is a circular graph in which each circle represents a possible state or remainder. The connections in the graph represent the operations performed during the calculation.

- Gray arrows represent the operation ' $r \rightarrow (10 \cdot r) \bmod m$ ', that is, they show what happens to the remainder when it is multiplied by 10.
- Circular connections represent the operation of adding one. They are used to visualize how the digit currently being processed is incorporated.
- Blue arrows show the step ' $r \rightarrow (10 \cdot r) \bmod m$ ' during the animation. They represent the stage in which the remainder obtained up to that point is multiplied by 10.
- Red arrows represent the walk followed while processing the previous digits. They make it possible to visually follow the walk taken by the number through the graph.
- The red node indicates the state in which the algorithm is located during the stage in which the current digit is added.

- The blue node indicates the state reached after multiplying the remainder obtained up to that point by 10.
- The final node appears in green when the calculation is completed. This node makes it possible to quickly identify the final remainder.

The entered number is displayed on the screen with its digits shown separately. During the animation, the digit currently being processed is highlighted. For example, if the number is 895958 and the third digit is being processed, it will appear as:

8 9 **5** 9 5 8

This makes it possible to identify at all times which part of the number is being processed by the algorithm. Once the calculation is complete, the highlighting disappears.

Below the number, the calculation being performed is displayed. For each digit, two operations are carried out:

1. Multiply by 10: $(10 \cdot r) \bmod m$.
2. Add the digit being processed: $(10 \cdot r + d) \bmod m$.

The resource displays these calculations together with their corresponding step number. At the same time, the «Information» box displays the sequence of states (or remainder walk) generated during the calculation. This makes it possible to observe how the remainder evolves without having to perform a traditional division of the entire number. Figure 1 shows a screenshot of the resource while the digit 5 of the number 895958 is being processed using the divisibility graph modulo 7.

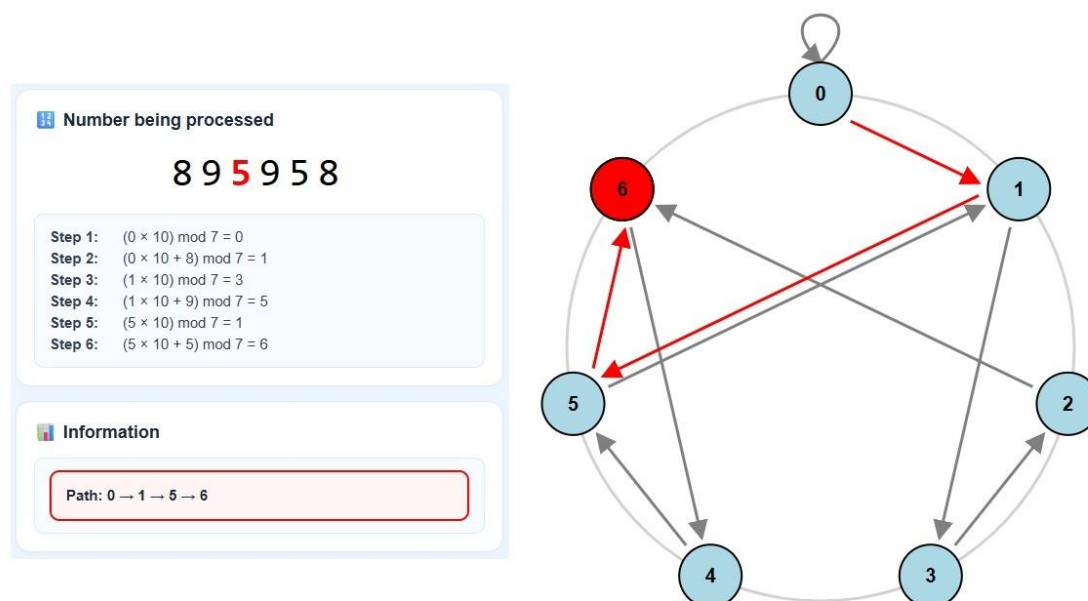


Figure 1. Resource during the simulation.

FINAL RESULT

Once all the steps have been completed, the final remainder, which is highlighted in green, must be examined. If the final remainder is 0, the number is divisible by the modulus. In the example shown in Figure 2, the sequence of states ends at 0. Therefore, the final remainder is 0 and, consequently, 895958 is divisible by 7.

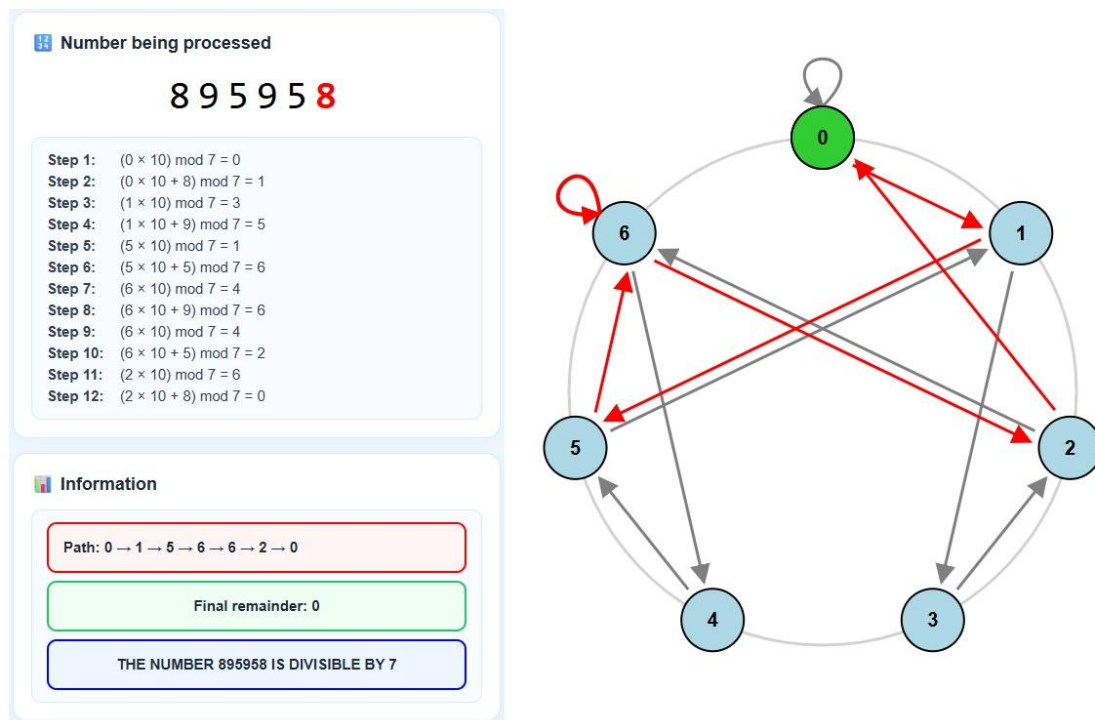


Figure 2. Final result of the simulation.

To start a new simulation with a different modulus and number, click the «New simulation» button. When this button is pressed, all the fields are cleared.

AUTOMATIC AND MANUAL MODES

The *automatic mode* allows users to observe the entire procedure without having to advance through the steps manually. Once the modulus and number have been entered and the speed has been adjusted, click the «Start automatic simulation» button. The user can then observe how the walk moves through the graph. During the automatic simulation, the «Pause» button can be used to stop the animation. When pressed, the button changes to «Resume», allowing the user to continue the animation from the point at which it was paused. There is no need to start again from the beginning.

The *manual mode* allows the user to control the process step by step. In this mode, the step slider is used. For example, if the slider displays «Step 6 / 12», this means that 6 of the 12 available steps have been completed. As explained above, two steps are performed for each digit: (1) multiply the current remainder by 10; and (2) add the digit being processed.

Therefore, if the number contains 'x' digits, there will be '2x' steps. To work in manual mode, once the modulus and number have been entered, click the «Start manual simulation» button and move the step slider to observe how the graph changes at each step. You can move both forward and backward through the process using the slider.